BIOLOGY TESTING QUESTION BANK

CBSE Class XII — Curated Question Bank from 6 Board Papers

Includes MCQs, Assertion–Reasoning, VSAQs, SAQs, LAQs and Case Study Questions.
Diagrams, graphs, tables, sequences and process figures are included wherever relevant.

Question Type	Count
MCQ	10
Assertion–Reasoning	6
VSAQ	10
SAQ	10
LAQ	10
Case Study	6
Total	52

SECTION A — MCQs

Each question carries 1 mark. Select the most appropriate option.

Q1. Identify the plant in which emasculation is not required for artificial hybridization.
(A) Rice
(B) Wheat
(C) Pea
(D) Papaya

Q2. Which of the following is not a functional unit of an ecosystem?
(A) Energy flow
(B) Decomposition
(C) Stratification
(D) Productivity

Q3. Identify the statements that correctly describe Darwin’s Theory of Evolution.
(i) Overproduction of organisms leads to competition.
(ii) Variation is inherited and causes evolution.
(iii) Acquired characters are inherited.
(iv) Survival depends on favourable traits.
(v) New species arise due to accumulation of favourable variations.
Choose the correct option.
(A) (i), (ii), (iv) and (v) are correct
(B) (ii), (iii) and (iv) are correct
(C) All are correct
(D) Only (i) and (iv) are correct

Q4. Given below are two columns. In Column I is the list of four enzymes and in Column II is the list of functions of the given enzymes. Which one of the following options shows the enzymes matched with their respective functions correctly?

Column I (Enzyme)
P. DNA Ligase
Q. Restriction exonuclease
R. Taq polymerase
S. Restriction endonuclease

Column II (Function)
i. Removes nucleotides from ends of DNA
ii. Extends primer on a DNA template
iii. Joins the DNA fragments
iv. Cuts DNA at a specific position

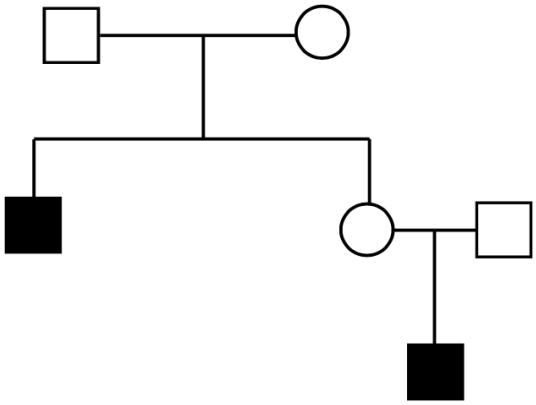
(a) P-i, Q-ii, R-iv, S-iii
(b) P-iv, Q-iii, R-ii, S-i
(c) P-i, Q-iv, R-iii, S-ii
(d) P-iii, Q-i, R-ii, S-iv

Q5. You know that there are twenty different types of naturally occurring amino acids and four different types of bases in the DNA. A combination of 3 such bases codes for a specific amino acid. If instead there are 96 different amino acids and 12 different bases in the DNA, then the minimum number of combination of bases required to form a codon is:
(A) 6
(B) 8
(C) 2
(D) 4

Q6. The rate of formation of new organic matter by consumers, and the biomass available for consumption of herbivores as well as decomposers are referred to as:
(A) Gross primary productivity and Net primary productivity
(B) Net primary productivity and Secondary productivity
(C) Secondary productivity and Net primary productivity
(D) Secondary productivity and Gross primary productivity

Q7. Given below are a few statements with reference to the accessory ducts of the human male reproductive system:
(i) The seminiferous tubules of the testes open into rete testis then into the vas deferens.
(ii) The vasa efferentia leave the testes and open into the epididymis.
(iii) The epididymis leads to vas deferens that ascends into the abdomen.
(iv) The vas deferens receives a duct from the prostate gland and opens into the urethra as ejaculatory duct.
(v) The urethra originates from the urinary bladder and extends through the penis to its external opening, urethral meatus.
Choose the option with all true statements from the given options:
(A) (i), (ii), (iv)
(B) (ii), (iii), (v)
(C) (ii), (iv), (v)
(D) (i), (iii), (iv)

Q8. In the given pedigree chart, a cross between a normal couple resulted in a son who was haemophilic and a normal daughter. In course of time, when the daughter was married to a normal man, to their surprise the grandson was also haemophilic. Choose the option that indicates the correct inheritance of trait in the above pedigree chart.
(A) Autosome linked dominant trait
(B) Sex-linked dominant trait
(C) Autosomal recessive trait
(D) Sex-linked recessive trait



Q9. Use the given information to select the amino acid attached to the 3′ end of tRNA during the process of translation, if the coding strand of the structural gene being transcribed has the nucleotide sequence TAC.
Codons for the amino acids:
AUC — Isoleucine
AUG — Methionine
UAC — Tyrosine
GUA — Valine
Options:
(A) Isoleucine
(B) Methionine
(C) Tyrosine
(D) Valine

Q10. In a pea plant (Pisum sativum) inflated pod shape is dominant over constricted pod shape. The expected ratio of phenotypes of the offspring in a cross between both the parents with heterozygous inflated pod shape will be:
(A) 1 : 0
(B) 1 : 1
(C) 2 : 1
(D) 3 : 1

SECTION A — Assertion–Reasoning

Each question carries 1 mark. For each question choose the correct option: (A) Both A and R are true and R is the correct explanation of A. (B) Both A and R are true but R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.

Q11. Assertion (A): The population of melanized moths increased in industrial areas after Industrial Revolution.
Reason (R): In industrial environment lichen-covered trees were replaced by soot-covered trees offering better camouflage to dark coloured moths.

Q12. Assertion (A): Linked genes do not show dihybrid F_2 ratio 9 : 3 : 3 : 1.
Reason (R): Linked genes do not undergo independent assortment.

Q13. Assertion (A): A cattle egret and grazing cattle in close association is a classic example of commensalism.
Reason (R): As grazing cattle move through the field, they stir up and flush out insects from the vegetation that otherwise might be difficult for egrets to find and catch.

Q14. Assertion (A): One of the properties of genetic code is degeneracy.
Reason (R): Some amino acids can be coded by more than one codon.

Q15. Assertion (A): A bioreactor provides the optimal conditions for achieving the desired product by providing optimum growth conditions.
Reason (R): The most commonly used bioreactors are of stirring type.

Q16. Assertion (A): A person infected with malaria suffers from chill and high fever, recurring every three or four days.
Reason (R): The parasite attacks the RBCs resulting in their rupture and release of haemozoin.

SECTION B — Very Short Answer Questions (VSAQs)

Each question carries 2 marks. Answer briefly.

Q17. (a) Define humoral immune response.
(b) Name any two types of antibodies found to give humoral immune response in humans.

Q18. (a) (i) How does a farmer use dormancy of seed to his advantage?
(ii) Differentiate between pea seed and castor seed.
OR
(b) Identify the stage of follicle where primary oocyte undergoes first meiotic division. Also mention the products of this division.

Q19. What did Henking observe in his experiments on insects that led to the discovery of sex chromosome? How did his observation contribute to our understanding of sex determination?

Q20. Write the basic steps followed in the Assisted Reproductive Technologies (ART) programme to help childless couples. Why is it also known as test tube baby programme?

Q21. (a) A farmer while working on his farm was bitten by a poisonous snake. He was rushed to a nearby health centre where the doctor gave him an injection to save his life.
(i) What did the doctor inject and why?
(ii) Name the kind of immunity provided by this injection.
OR
(b) Why do organic farmers not recommend complete eradication of insect pests? Explain giving reason.

Q22. The allele for brown eyes (B) is dominant over blue eyes (b). In a population, 36% of individuals have blue eyes. Assuming Hardy–Weinberg equilibrium, calculate:
(i) Frequency of allele B
(ii) Percentage of heterozygous individuals
(iii) Percentage of individuals that are homozygous dominant
(Working to be shown)

Q23. (a) Explain how the immunity of a person is affected if there is atrophy/degradation of the thymus gland at an early stage of life.
OR
(b) (i) What are interferons? Explain their role in providing immunity to a person.
(ii) Which category of innate immunity defence barrier can interferons be classified into?

Q24. Assume that the given mRNA start site is not depicted and is theoretically translated in two reading frames.
(a) Translation starting from the first nucleotide — Reading frame 1
5′ CUCGCUUGCCGAUCAAGGGUUA 3′
(b) Translation starting from the second nucleotide — Reading frame 2

5′ GUGGCACUCAGUCCUUAAUGGCG 3′
How many amino acids will be specified in case (a) and case (b) on translation? Justify your answer.

Q25. The basic scheme of the essential steps involved in the process of recombinant DNA technology is summarized below in the form of a flow diagram. Study the given flow diagram and answer the questions that follow.
(a) Name the enzyme used in Step-1 to join the cut plasmid and alien DNA.
(b) State the technical term used for Step-3.
(c) Justify the use of the same restriction enzyme EcoRI to cut both the vector DNA and the alien DNA.

Step-1: Vector DNA cut using restriction enzyme EcoRI + alien DNA cut using restriction enzyme EcoR
↓ Enzyme A
Step-2: Recombinant DNA molecule
↓
Step-3: Transfer of recombinant DNA molecule in E. coli host
↓
Step-4: Replication of recombinant DNA molecule in E. coli

Q26. (a) Explain how the interaction between sea anemone and clownfish is one of the best examples of commensalism in nature.
OR
(b) Correctly depict, also indicate the trophic level and describe the ecological pyramid of biomass in sea with 40 standing crop of phytoplankton supporting 90 standing crop of zooplankton which further supports 120 small fishes.

SECTION C — Short Answer Questions (SAQs)

Each question carries 3 marks. Answer with the necessary details.

Q27. Explain the mode of action of contraceptive pills taken by human females. Mention the schedule to be followed for effective outcome.

Q28. (a) What are transgenic animals?
(b) Name the first transgenic cow and state its importance.
OR
(a) Explain the convention for naming EcoRI.
(b) With the help of an illustration only, show the action of EcoRI on a DNA polynucleotide.

Q29. (a) What is the carrying capacity of a species in a habitat?
(b) Explain the growth curve that takes this capacity into account.

Q30. (a) Explain the different ways by which apomictic seeds can be developed.
(b) Mention one advantage of apomictic seeds for farmers.

Q31. (a) What is meant by translation in protein synthesis?
(b) Explain charging of tRNA, aminoacylation of tRNA, and mention its importance in the process of translation.

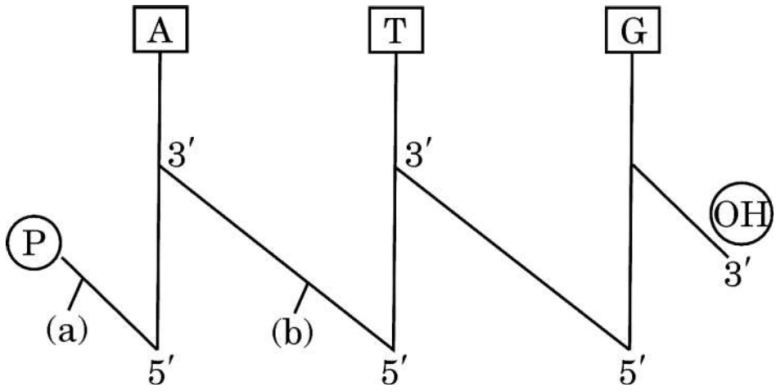
Q32. Identify A, B, C, D, E and F in the table given below:

Name of Human Disease	*Causative Organism*	*Symptoms*
Pneumonia	*Streptococcus*	'A'
Typhoid	'B'	High fever, weakness, headache, stomach pain
Common Cold	Rhino virus	'C'
Ringworm	'D'	Dry scaly lesions on body parts, redness, itching
Ascariasis	*Ascaris*	'E'
'F'	*Entamoeba histolytica*	Constipation, cramps, stools with mucous and blood clots

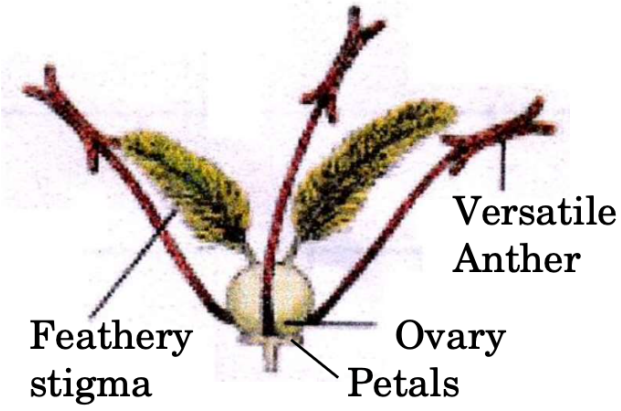
Q33. With the help of a schematic diagram only, show in three steps, the formation of recombinant DNA by the action of restriction endonuclease EcoRI and DNA ligase.

Q34. Study the single strand of deoxyribonucleic acid depicted in the stick diagram and answer the questions that follow.
(a) Name the covalent bonds depicted as (a) and (b) in the form of slanting lines in the diagram.

(b) How many purines are present in the given strand?
(c) Draw the chemical structure of the given polynucleotide chain of DNA.



Q35. Given below is a flower with its characteristic features specialised for the most common type of abiotic pollination. Answer the following questions based on the diagram.
(a) Name the mode of abiotic pollination that will be adopted by the given plant species in the above picture.
(b) State the need of exposed large feathery stigmas for the flower.
(c) What will be the two important adaptations in the pollen grains of the flowers pollinated by the above mode of pollination?
(d) What could be the probable reason for the petals being small and non-green?



Q36. (a) Describe the experiment conducted by T. H. Morgan on Drosophila melanogaster involving eye colour and body colour.
(b) How did the results deviate from Mendelian inheritance pattern?
(c) Explain the two genetic terms used by Morgan for his observations.

SECTION E — Long Answer Questions (LAQs)

Each question carries 5 marks. Answer with detailed explanation wherever required.

Q37. Trace the events from fertilization till implantation of blastocyst in human female. Also mention the site where fertilization takes place.

Q38. (a) (i) Watson and Crick's discovery of double helical structure was based on which two findings. Also mention the names of the scientists associated with these findings.
(ii) Write down the salient features of double helix structure of DNA (any three points).
OR
(b) Given below is a stretch of DNA showing the coding strand of structural gene of transcription unit.
5′ - ATG ACC GTA TTT TCT GTA GTG CCC GTA CTT CAG GCA TTA - 3′
(i) Write the corresponding template strand and mRNA strand that will be transcribed along with its polarity.
(ii) If GUA of transcribed mRNA is an intron, then depict the sequence involved in formation of mRNA / mature / processed hnRNA strand: (1) In a bacterium (2) In humans.
(iii) How many amino acids will the resulting polypeptide have after the process of translation in humans?

Q39. (a) (i) Name the protozoan species that is responsible for causing the most serious and even fatal malarial disease.
(ii) Name the host in which the parasite completes its sexual stages and explain the changes taking place.
(iii) How does the parasite damage the human body after entering the blood stream?
(iv) Suggest two effective preventive measures to control the spread of this disease in endemic regions.
OR
(b) (i) What are bio-fertilizers?
(ii) Name the different types of microorganisms used as bio-fertilizers in organic farming and explain how each contributes to soil fertility.
(iii) Write two advantages of using bio-fertilizers over chemical fertilizers.

Q40. (a) (i) Describe the series of experiments conducted by Frederick Griffith. Comment on the significance of the result obtained.
(ii) State the contribution of Avery, MacLeod and McCarty.
OR
(b) The following questions are based on haemophilia.
(i) Work out the crosses between: (I) Normal female and haemophilic male, (II) Carrier female and normal male, (III) Carrier female and haemophilic male.
(ii) Write the conclusions you draw from these crosses. Comment on the type of inheritance of the disease.
Use: X — Normal, X^h — Haemophilic.

Q41. (a) (i) Explain any two basic principles/core techniques on which biotechnology is based.
(ii) Describe any three key tools used in recombinant DNA technology.

OR
(b) “Early and accurate diagnosis of diseases is vital in medical technology.”
(i) Name the conventional methods of diagnosis and their disadvantages.
(ii) Which three diagnostic techniques have been developed through Biotechnology? Explain how each one helps in detecting diseases.

Q42. (a) (i) Explain how double fertilisation takes place in a flowering plant.
(ii) Write the fate of the products of double fertilization in these plants.
OR
(b) (i) Explain the structure of testicular lobules in human male reproductive system. Name the two types of cells present in the seminiferous tubules and state their role.
(ii) Describe the role of hypothalamic hormone GnRH in spermatogenesis.

Q43. (a) Meselson and Stahl carried out an experiment to prove the nature of DNA replication. Recall the experiment and answer the following questions.
(i) Which two types of nitrogen were used by them in their experiment and why?
(ii) Why did they take samples of E. coli at definite time intervals for their observation?
(iii) State the role of caesium chloride density gradient in their experiment.
(iv) Write the conclusions they arrived at.
OR
(b) A true-breeding pea plant with round seeds and tall plant was crossed with a pea plant having wrinkled seeds and dwarf plant. Represent this cross up to F_2 generation showing F_1 and F_2 phenotypic ratio respectively.
(ii) State the Mendel’s law that can be derived from this type of cross.

Q44. (a) Name and explain the biotechnological strategy wherein the infection by the nematode Meloidogyne incognita can be prevented using Agrobacterium vectors in the roots of tobacco plant by RNA interference.
OR
(b) Explain the amplification of gene of interest using the technique of Polymerase Chain Reaction (PCR).

Q45. (a) (i) Describe the population growth curve applicable in a population of any species in nature that has limited resources at its disposal.
(ii) Give the equation of this growth curve.
(iii) Name the growth curve and depict a graphical plot for this type of population growth.
OR
(b) (i) Explain the Species–Area relationship within a natural forest and also predict the nature of graph when species richness is plotted against the area for a wide variety of taxa.
(ii) Depict the graphical relationship between species richness and area.

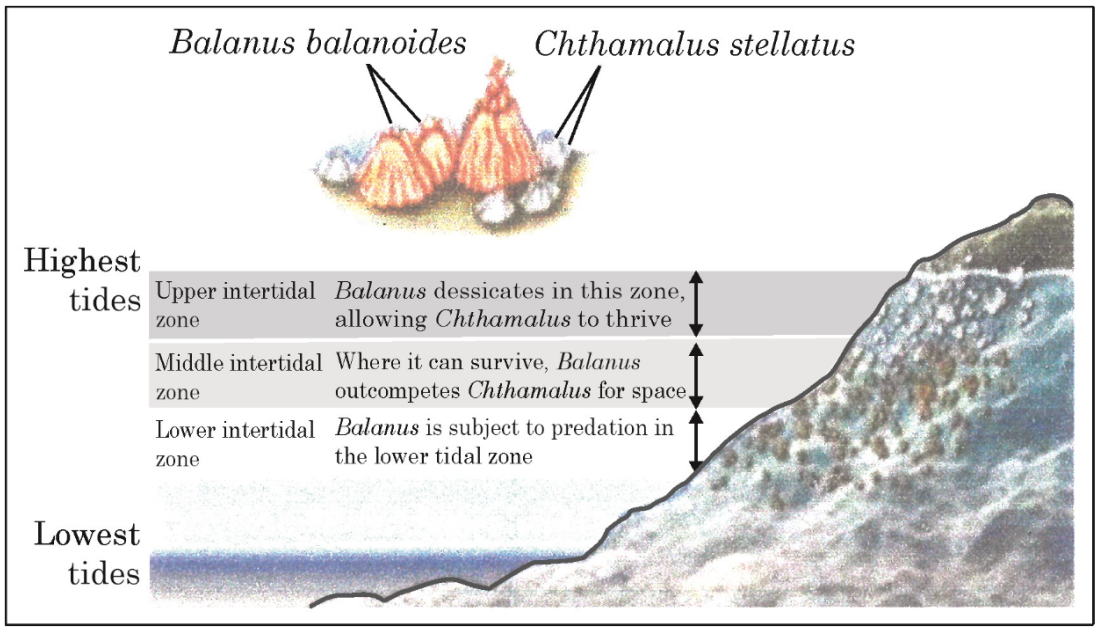
Q46. Stability, as one of the properties of genetic material, was very evident in one of the very early experiments in genetics. Name the scientist and describe the experiment. State the conclusion arrived at.
OR
A tall pea plant bearing violet flowers with unknown genotype is given. Find the genotype by crossing it with a suitable plant. Work out the cross to support your

answer.

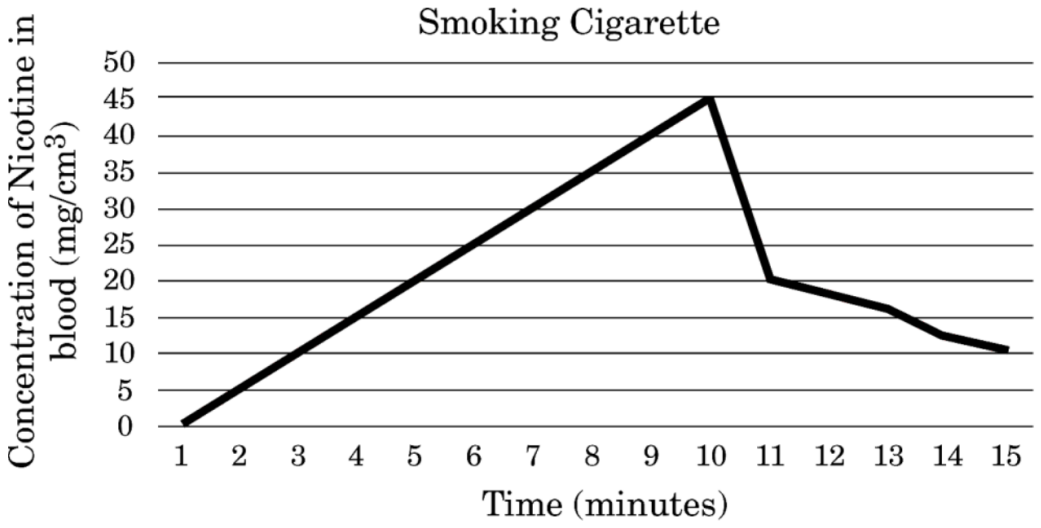
SECTION D — Case Study Questions

Each question carries 4 marks. Read the passage / study the figure carefully and answer the sub-questions.

Q47. The diagram below shows the distribution of two barnacle species, Chthamalus and Balanus on a rocky sea shore. When Balanus is experimentally removed, Chthamalus expands its range in lower intertidal zone.
(a) Identify and define the ecological phenomenon demonstrated by this observation.
OR
(a) State the principle that explains the elimination of one species.
(b) Two different species can compete for the same resource. Give another example of it.
(c) How do species avoid competition in nature? Explain with an example.



Q48. The graph below shows the concentration of nicotine in blood of a smoker taking 10 puffs per minute.
(a) (i) With reference to the above graph, explain the concentration of nicotine in the blood at 10 minutes.
OR
(a) (ii) How will this affect the concentration of carbon monoxide and haem-bound oxygen at 10 minutes?
(b) How does cigarette smoking result in high blood pressure and increase in heart rate?
(c) To which class of compounds does nicotine belong? Name one other drug from the same class.



Q49. Read the following passage and answer the questions that follow. India's land area is only a small percentage of the world's total land area whereas its share of the global species diversity is an impressive 8.1 per cent. However, in these estimates of species, prokaryotes do not figure anywhere. Biologists are always keen on collecting data with respect to species diversity observed in different regions of the world. The data collected based on the survey conducted for species richness of groups of mammals in three different regions of the world is shown in the bar graph given below.

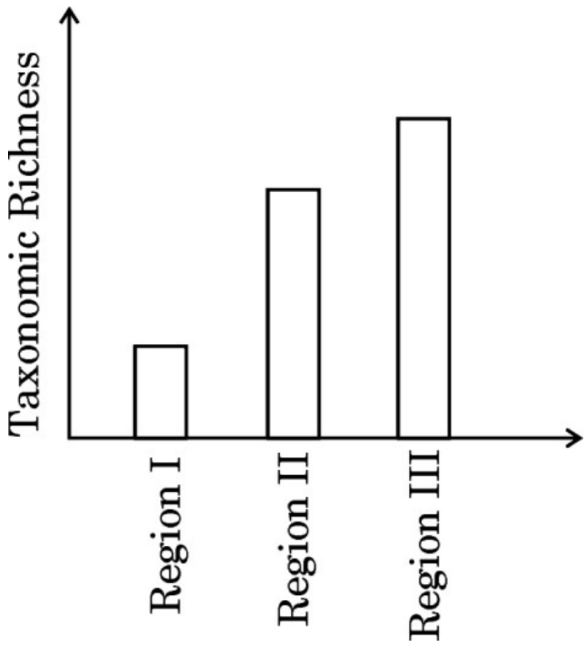
(a) Why is the species richness maximum in Region III in the bar graph?

OR

(a) Why is the species richness minimum in Region I in the bar graph?

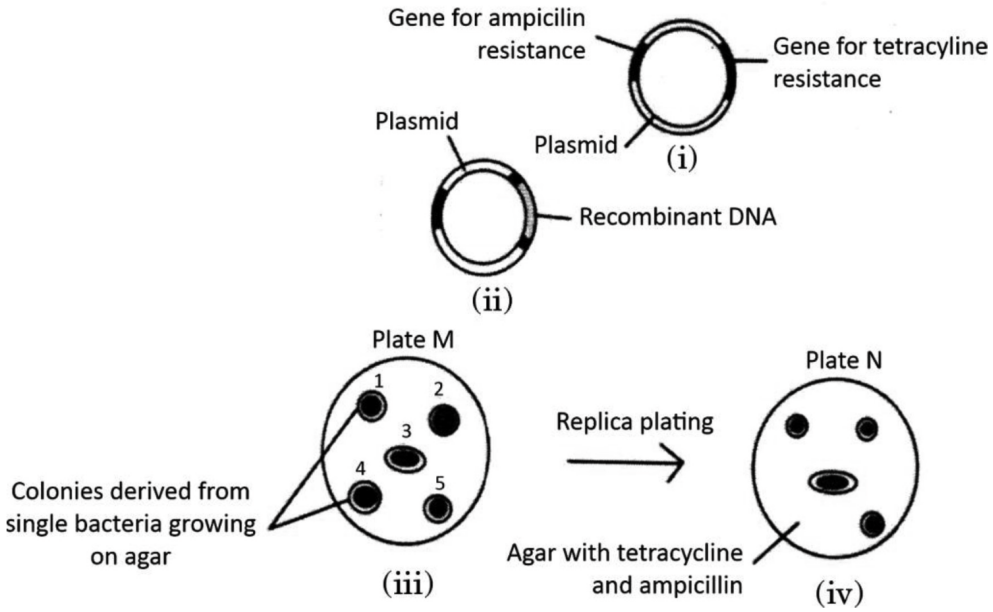
(b) Plants and animals do not have uniform diversity in the world but show rather uneven distribution. Mention what this kind of diversity is referred to as.

(c) Why is it that prokaryotes do not have an estimated number of their species diversity as seen in plants and animals? Explain.



Q50. Read the following passage and answer the questions that follow. Gene expresses itself in a cell system as a protein/enzyme. How does expression of gene occur in a cell system and when does it need to occur, and how gene expression is regulated in a prokaryote cell system, was studied by the combined efforts of Jacques Monod, the biochemist, and François Jacob, the geneticist. For their work on lactose metabolism in E. coli and proposing the lac operon concept, they were awarded the Nobel Prize in 1965.
(a) Why is lac operon said to be a transcriptionally regulated system?
(b) It is said that the lac operon has to be operational at a very low level in the bacterial cell all the time. Justify.
(c) Why is the regulator gene in lac operon called i gene?
(d) Draw a schematic diagram of lac operon in absence of inducer in the culture medium of the bacteria.
OR
(d) Draw a schematic diagram of lac operon in the presence of inducer in the culture medium of the bacteria.

Q51. Study the diagram that shows the steps involved in the procedure of selecting transformed bacteria and answer the questions that follow.
(a) Identify the colony that has got transformed. Justify your answer.
(b) What are the sites in a plasmid called where ampicillin and tetracycline resistance genes are inserted? State their role in genetic engineering.
(c) Name two enzymes playing an important role in genetic engineering.
OR
(c) State the role of β-galactosidase in insertional inactivation.



Q52. Read the following passage and answer the questions that follow. The most convincing evidence to trace evolutionary relationships between humans and different groups of animals comes from the basic similarities seen at the molecular level. Study the table below that depicts the number of amino acid differences between the haemoglobin polypeptide of a few animals with that of humans and answer the

questions that follow.
(a) To which category of evolution — divergent or convergent — do the following evolutionary relationships belong to: (i) Humans and Macaque, (ii) Humans and Frog?
(b) What do the biochemical similarities in haemoglobin suggest about the evolutionary relationship between humans, frog and lamprey?
(c) (i) Which one of the two is closely related to humans and why?
OR
(c) (ii) Which one of the two is distantly related to humans and why?

S.No.	Animal	No. of amino acid differences in haemoglobin with that of humans
1	Macaque	08
2	Dog	32
3	Bird	45
4	Frog	67
5	Lamprey	125